Population-Scaled Capital and Leisure: Dynamic, Stochastic, and Machine-Learned Equilibrium
Expanded Version

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Abstract

We construct a competitive macroeconomic model in which both the capital stock and aggregate leisure scale proportionally with population, yielding a per-capita stationary equilibrium that is independent of population size. We derive a closed-form interior solution for the equilibrium leisure-consumption allocation and establish its uniqueness and efficiency properties under standard regularity conditions. The framework is extended along four dimensions. First, we formulate a continuous-time Ramsey optimal growth model with population growth and depreciation, deriving the Euler equation and transversality conditions rigorously. Second, we introduce geometric Brownian motion productivity shocks, establishing the stochastic Hamilton–Jacobi–Bellman (HJB) equation and its viscosity solution. Third, we conduct a rigorous spectral analysis of the Bellman operator, deriving Fréchet derivatives, resolvent structure, and Witten-type spectral asymptotics that characterize macroeconomic regime-switching timescales. Fourth, we develop a neural PDE solver for the HJB equation and a machine-learning surrogate for equilibrium computation, providing convergence guarantees and error bounds. A multi-country network extension characterizes global stability via the eigenvalues of a composite Jacobian. Taken together, these results bridge classical Ramsey theory, stochastic control, spectral geometry, and modern computational methods within a unified population-scaled framework.

The paper ends with "The End"

Keywords: Ramsey growth, population scaling, Hamilton–Jacobi–Bellman equation, spectral analysis, Witten Laplacian, Morse theory, neural PDE, machine learning surrogate, regime switching, multi-country equilibrium.

Contents

1 Introduction **2**

2 Population-Scaled Economy: Static Foundations **3**
- 2.1 Technology and Aggregation . . . 3
- 2.2 Households . . . 3
- 2.3 Firms and Factor Prices . . . 3
- 2.4 Equilibrium Leisure and Consumption . . . 4
- 2.5 Efficiency and the First Welfare Theorem . . . 4

3 Dynamic Ramsey Model **5**
- 3.1 Capital Accumulation and the Planner's Problem . . . 5
- 3.2 Hamiltonian and Optimality Conditions . . . 5
- 3.3 Steady State and Phase Diagram Analysis . . . 5

4 Stochastic Productivity **6**
- 4.1 Geometric Brownian Motion for TFP . . . 6
- 4.2 Stochastic Capital Dynamics . . . 6

5 Stochastic Dynamic Programming **7**
- 5.1 The Value Function and HJB Equation . . . 7
- 5.2 Optimality Conditions and Policy Functions . . . 7

6 Spectral Analysis of the Equilibrium Operator **7**
6.1 The Bellman Operator . . . 7
6.2 Fréchet Derivative of the Bellman Operator . . . 8

7 Resolvent and Spectral Gap Analysis **8**
7.1 Generator and Resolvent . . . 8

8 Diffusion Geometry and the Witten Laplacian **8**
8.1 Potential Landscape and Witten Transform . . . 8

9 Morse Landscape of Macroeconomic Equilibria **9**
9.1 Critical Point Structure . . . 9

10 Spectral Asymptotics and Metastable Regimes **9**
10.1 The Setting . . . 9
10.2 Witten-Type Spectral Asymptotics . . . 9
10.3 Economic Interpretation . . . 10
10.3.1 Regime-Switching Timescales . . . 10
10.3.2 Interpretation of Regimes . . . 10
10.3.3 Return-Time Distributions . . . 10

11 Neural PDE Solver for the HJB Equation **10**
11.1 Motivation and Architecture . . . 10
11.2 Loss Function and Training . . . 11

12 Econometric Estimation **11**
12.1 Identification Strategy . . . 11
12.2 Moment Conditions . . . 12

13 Machine Learning Surrogate Model **12**
13.1 Motivation . . . 12
13.2 Surrogate Architecture and Training . . . 12

14 Multi-Country Network Equilibrium **13**
14.1 Model with International Capital Flows . . . 13

15 Conclusion **13**
15.1 Future Directions . . . 14

A Proof of the Intratemporal Optimality Condition **14**

B References **14**

C Glossary of Key Notation **15**

List of Tables

1 Structural parameter identification strategy. . . . 12

1 Introduction

A central challenge in macroeconomic theory is reconciling aggregate behavioral relationships with the underlying structure of individual decision-making in economies of varying size. Standard representative-agent models sidestep this issue by working directly in per-capita terms, but doing so requires explicit justification of the scaling assumptions that permit aggregation.

This paper takes these scaling assumptions seriously. We posit that aggregate capital K and aggregate leisure L scale *linearly* with population N—i.e., $K = kN$ and $L = \ell N$—and show that this structural assumption alone is sufficient to reduce all equilibrium conditions to a system in per-capita variables (c, k, ℓ) that is *independent of the level of* N. This is not merely a normalization; it reflects a deep homogeneity

property of the Cobb–Douglas production technology and log-separable preferences that we make precise in Section 2.

Starting from this foundation, we pursue four interconnected extensions. The **dynamic extension** (Section 3) embeds the static model in a Ramsey optimal growth framework, deriving the continuous-time Euler equation and establishing the saddle-point stability of the steady state. The **stochastic extension** (Sections 4–5) introduces log-normal productivity shocks via geometric Brownian motion, formulating the stochastic dynamic programming problem and characterizing the value function as a viscosity solution to the HJB equation. The **spectral-geometric extension** (Sections 6–10) conducts a functional-analytic study of the Bellman operator, connecting the equilibrium dynamics to Witten-type Laplacians and Morse theory. Finally, the **computational extension** (Sections 11–13) introduces neural PDE solvers and ML surrogate models, complete with training algorithms and error analyses.

Throughout, we maintain a parallel track of *economic interpretation*: each mathematical result is accompanied by a discussion of what it means for macroeconomic dynamics, policy, and empirical estimation.

2 Population-Scaled Economy: Static Foundations

2.1 Technology and Aggregation

Let $N > 0$ denote the size of a representative population. Each agent is endowed with one unit of time per period, which they allocate between labor $1-\ell$ and leisure $\ell \in (0,1)$. Aggregate capital and aggregate leisure supplied to the market satisfy:

$$K = kN, \qquad L = \ell N, \tag{1}$$

where $k > 0$ is per-capita capital and $\ell \in (0,1)$ is per-capita leisure. Aggregate labor supply is therefore $N(1-\ell)$.

The single representative firm operates a Cobb–Douglas production technology:

$$Y = AK^{\alpha}\big(N(1-\ell)\big)^{1-\alpha}, \qquad \alpha \in (0,1), \tag{2}$$

where $A > 0$ is total factor productivity (TFP) and α is the capital income share. Substituting (1):

$$Y = A(kN)^{\alpha}\big(N(1-\ell)\big)^{1-\alpha} = Ak^{\alpha}(1-\ell)^{1-\alpha}N^{\alpha+1-\alpha} = Ak^{\alpha}(1-\ell)^{1-\alpha}N. \tag{3}$$

Dividing by N, per-capita output is:

$$\boxed{y = Ak^{\alpha}(1-\ell)^{1-\alpha}.} \tag{4}$$

Remark 2.1. *The linear homogeneity of Y in (K,N) is essential for (4). The Cobb–Douglas form with exponents summing to one (constant returns to scale) ensures that the scaling (1) collapses the N-dependence exactly. Extensions to increasing returns would break this property and reintroduce population-level effects.*

2.2 Households

A continuum of identical households of measure N each maximizes the strictly concave, time-separable utility function:

$$U(c,\ell) = \log c + \theta \log \ell, \qquad \theta > 0, \tag{5}$$

where $c > 0$ is per-capita consumption and θ parametrizes the relative utility weight on leisure. The additively separable log specification implies unit income elasticity for both goods and guarantees an interior optimum under the Inada-type conditions $U_c \to \infty$ as $c \to 0$ and $U_\ell \to \infty$ as $\ell \to 0$.

Each household earns labor income $w(1-\ell)$ and capital income rk, facing the per-capita budget constraint:

$$c = w(1-\ell) + rk. \tag{6}$$

2.3 Firms and Factor Prices

Competitive firms maximize profits by choosing K and L, taking factor prices (r,w) as given:

$$\max_{K,L}\ AK^{\alpha}L^{1-\alpha} - rK - wL. \tag{7}$$

First-order conditions (Inada conditions ensure interior solutions):

$$w = (1-\alpha)AK^{\alpha}L^{-\alpha} = (1-\alpha)Ak^{\alpha}(1-\ell)^{-\alpha}, \tag{8}$$

$$r = \alpha AK^{\alpha-1}L^{1-\alpha} = \alpha Ak^{\alpha-1}(1-\ell)^{1-\alpha}. \tag{9}$$

Economic Intuition. Equation (8) states that the real wage equals labor's marginal product: as workers are more productive (higher A) or better capitalized (higher k), wages rise. Equation (9) is the analogous statement for capital: the rental rate is the marginal product of capital, which diminishes in k (consistent with decreasing returns to the individual factor).

2.4 Equilibrium Leisure and Consumption

Assumption 2.1. *Parameters satisfy $\theta > 0$, $\alpha \in (0,1)$, $A > 0$, $k > 0$. These ensure interior solutions throughout.*

The household maximizes (5) subject to (6), treating w and r as given. Because goods-market clearing requires $c = y = Ak^{\alpha}(1-\ell)^{1-\alpha}$, the budget constraint is automatically satisfied and the problem reduces to choosing ℓ to maximize $\log[Ak^{\alpha}(1-\ell)^{1-\alpha}] + \theta \log \ell$. The first-order condition yields the intratemporal optimality condition:

$$\frac{\theta}{\ell} = \frac{w}{c}. \tag{10}$$

This is the familiar condition: the marginal rate of substitution between leisure and consumption equals the real wage (the opportunity cost of leisure). Substituting (8) and $c = Ak^{\alpha}(1-\ell)^{1-\alpha}$:

$$\frac{\theta}{\ell} = \frac{(1-\alpha)Ak^{\alpha}(1-\ell)^{-\alpha}}{Ak^{\alpha}(1-\ell)^{1-\alpha}} = \frac{1-\alpha}{1-\ell}. \tag{11}$$

Cross-multiplying and solving:

$$\theta(1-\ell) = \ell(1-\alpha) \implies \theta - \theta\ell = \ell(1-\alpha) \implies \theta = \ell(\theta + 1 - \alpha). \tag{12}$$

$$\boxed{\ell^* = \frac{\theta}{\theta + 1 - \alpha}.} \tag{13}$$

Proposition 2.1. *Under Assumption 2.1, the static competitive equilibrium admits a unique interior solution (ℓ^*, c^*, y^*) given by:*

$$\ell^* = \frac{\theta}{\theta + 1 - \alpha}, \tag{14}$$

$$y^* = Ak^{\alpha}(1-\ell^*)^{1-\alpha} = Ak^{\alpha}\left(\frac{1-\alpha}{\theta+1-\alpha}\right)^{1-\alpha}, \tag{15}$$

$$c^* = y^* \quad \textit{(goods market clearing).} \tag{16}$$

Proof. Existence and uniqueness of ℓ^* follow from the strict concavity of U in ℓ on $(0,1)$ and continuity of the first-order condition. The value $\ell^* \in (0,1)$ since $\theta > 0$ and $1-\alpha > 0$ imply $0 < \theta < \theta + 1 - \alpha$. Goods market clearing then pins down $c^* = y^*$. □

Economic Intuition. The closed-form solution (13) reveals three testable comparative statics: (i) $\partial\ell^*/\partial\theta > 0$: households with stronger leisure preferences work less; (ii) $\partial\ell^*/\partial\alpha > 0$: a higher capital share means labor is relatively less important in production, reducing the wage and thus the opportunity cost of leisure; (iii) ℓ^* is independent of A and k—productivity and capital do not affect the *fraction* of time devoted to leisure, only the level of output. This scale-neutrality is a direct consequence of the log utility specification.

2.5 Efficiency and the First Welfare Theorem

Proposition 2.2 (Efficiency)**.** *The competitive equilibrium allocation (ℓ^*, c^*) coincides with the solution to the social planner's problem:*

$$\max_{c,\ell} \ \log c + \theta \log \ell \quad s.t. \quad c = Ak^{\alpha}(1-\ell)^{1-\alpha}.$$

Proof. Substituting the constraint $c = Ak^{\alpha}(1-\ell)^{1-\alpha}$ into the objective, the planner maximizes $h(\ell) = (1-\alpha)\log(1-\ell) + \theta \log \ell$ over $\ell \in (0,1)$ (the terms involving A and k are constants with respect to ℓ). Differentiating: $h'(\ell) = -(1-\alpha)/(1-\ell) + \theta/\ell = 0$, which is identical to the household's first-order condition. Hence both problems yield the same optimum (13). □

3 Dynamic Ramsey Model

3.1 Capital Accumulation and the Planner's Problem

We embed the static framework in a continuous-time optimal growth model. Let $n \geq 0$ denote the exogenous rate of population growth and $\delta > 0$ the depreciation rate of capital. Per-capita capital evolves as:

$$\dot{k} = Ak^{\alpha}(1-\ell)^{1-\alpha} - c - (n+\delta)k, \tag{17}$$

where the term $(n+\delta)k$ accounts for both depreciation and the capital dilution effect of population growth (new workers enter without capital).

The benevolent social planner solves:

$$\max_{c(t),\ell(t)} \int_0^{\infty} e^{-\rho t}\big[\log c(t) + \theta \log \ell(t)\big]\, \mathrm{d}t, \tag{18}$$

subject to (17), the initial condition $k(0) = k_0 > 0$, and the non-negativity constraints $c(t), \ell(t) > 0$ for all $t \geq 0$.

Assumption 3.1 (Feasibility)**.** *The discount rate satisfies $\rho > n$. This ensures that the present-discounted value of utility remains finite along any feasible path with log utility, since per-capita consumption cannot grow faster than n in the long run under constant returns to scale with no exogenous technical progress.*

3.2 Hamiltonian and Optimality Conditions

The current-value Hamiltonian is:

$$\mathcal{H} = \log c + \theta \log \ell + \lambda\big[Ak^{\alpha}(1-\ell)^{1-\alpha} - c - (n+\delta)k\big], \tag{19}$$

where $\lambda(t) > 0$ is the co-state variable representing the shadow price of per-capita capital (the marginal utility of wealth).

First-order conditions:

$$\frac{\partial \mathcal{H}}{\partial c} = 0: \quad \frac{1}{c} = \lambda, \tag{20}$$

$$\frac{\partial \mathcal{H}}{\partial \ell} = 0: \quad \frac{\theta}{\ell} = \lambda(1-\alpha)Ak^{\alpha}(1-\ell)^{-\alpha} = \lambda w, \tag{21}$$

$$\dot{\lambda} = \rho\lambda - \frac{\partial \mathcal{H}}{\partial k}: \quad \dot{\lambda} = \lambda\big[\rho + n + \delta - \alpha A k^{\alpha-1}(1-\ell)^{1-\alpha}\big]. \tag{22}$$

Euler equation. Differentiating (20) with respect to time gives $\dot{\lambda} = -\dot{c}/c^2$. Substituting into (22) and using $\lambda = 1/c$:

$$\boxed{\frac{\dot{c}}{c} = \alpha A k^{\alpha-1}(1-\ell)^{1-\alpha} - (\rho + n + \delta) = r - (\rho + n + \delta),} \tag{23}$$

where $r = \alpha A k^{\alpha-1}(1-\ell)^{1-\alpha}$ is the marginal product of capital. Consumption grows when the return to capital exceeds the sum of the discount rate, population growth, and depreciation, and declines otherwise. The n term arises because capital must be spread over an ever-growing population: new workers enter without assets, diluting the per-capita capital stock.

Transversality condition:

$$\lim_{t\to\infty} e^{-\rho t}\lambda(t)k(t) = 0. \tag{24}$$

This rules out Ponzi schemes: the present value of wealth must vanish as $t \to \infty$.

3.3 Steady State and Phase Diagram Analysis

At the steady state (k^*, c^*, ℓ^*), we have $\dot{c} = 0$, $\dot{k} = 0$, and $\ell = \ell^*$ (from the static intratemporal condition, which continues to hold dynamically). Setting $\dot{c}/c = 0$ in (23) and $\dot{k} = 0$ in (17):

$$r^* = \alpha A (k^*)^{\alpha-1}(1-\ell^*)^{1-\alpha} = \rho + n + \delta, \tag{25}$$

$$c^* = A(k^*)^{\alpha}(1-\ell^*)^{1-\alpha} - (n+\delta)k^*. \tag{26}$$

Equation (25) is the *modified golden rule*: at the optimum, the marginal product of capital equals the effective cost of holding capital (the discount rate ρ, the population dilution rate n, and physical depreciation δ). Solving (25) for k^*:

$$k^* = \left[\frac{\alpha A(1-\ell^*)^{1-\alpha}}{\rho+n+\delta}\right]^{1/(1-\alpha)}. \tag{27}$$

Proposition 3.1 (Saddle-Path Stability)**.** *The steady state (k^*, c^*) is a saddle point of the dynamical system $(\dot{k}, \dot{c})$ defined by (17) and (23). The eigenvalues of the Jacobian evaluated at the steady state are real with opposite signs, implying a one-dimensional stable manifold.*

Proof. Treating $\ell = \ell^*$ as pinned by the static condition, the dynamical system in (k, c) is $\dot{k} = Ak^\alpha(1-\ell^*)^{1-\alpha} - c - (n+\delta)k$ and $\dot{c} = c[r(k) - (\rho+n+\delta)]$ where $r(k) = \alpha A k^{\alpha-1}(1-\ell^*)^{1-\alpha}$. Evaluating partial derivatives at (k^*, c^*) and using $r^* = \rho+n+\delta$:

$$J = \begin{pmatrix} r^* - (n+\delta) & -1 \\ c^*\,\alpha(\alpha-1)A(k^*)^{\alpha-2}(1-\ell^*)^{1-\alpha} & 0 \end{pmatrix} = \begin{pmatrix} \rho & -1 \\ c^*\,r^*(\alpha-1)/k^* & 0 \end{pmatrix}.$$

The determinant is $\det(J) = c^*\,r^*(\alpha-1)/k^* < 0$ since $\alpha \in (0,1)$. A negative determinant implies that the two eigenvalues are real with opposite signs, so the steady state is a saddle. The unique stable arm is the equilibrium path. □

Economic Intuition. Saddle-path stability is the hallmark of the Ramsey model: starting from any k_0, there is a unique consumption path that converges to the steady state. This "rational expectations" property selects the stable arm of the saddle as the equilibrium trajectory and rules out explosive or deflationary paths. The phase diagram illustrates that to the left of k^*, the capital stock is below its optimal level and consumption must be kept low to allow accumulation; to the right, the economy is overcapitalized and should run down its stock by consuming more. The steady-state capital k^* in (27) is decreasing in ρ (impatient societies accumulate less), decreasing in n (faster population growth requires capital dilution), and decreasing in δ (higher depreciation makes capital more costly to maintain).

4 Stochastic Productivity

4.1 Geometric Brownian Motion for TFP

Deterministic models miss the rich short-run fluctuations observed in macroeconomic data. We introduce stochastic total factor productivity modeled as a geometric Brownian motion:

$$\mathrm{d}A_t = \mu A_t\,\mathrm{d}t + \sigma A_t\,\mathrm{d}W_t, \qquad A_0 > 0, \tag{28}$$

where W_t is a standard Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$, $\mu \in \mathbb{R}$ is the drift (average growth rate of TFP), and $\sigma > 0$ governs the volatility of productivity shocks.

By Itô's lemma, $\log A_t$ follows an arithmetic Brownian motion:

$$\mathrm{d}\log A_t = \left(\mu - \tfrac{1}{2}\sigma^2\right)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t. \tag{29}$$

This implies $A_t = A_0 \exp\left[(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\right]$, so $A_t > 0$ almost surely for all t—preserving economic feasibility.

4.2 Stochastic Capital Dynamics

Per-capita output and capital dynamics become stochastic:

$$y_t = A_t k_t^\alpha (1-\ell_t)^{1-\alpha}, \tag{30}$$

$$\mathrm{d}k_t = \left[y_t - c_t - (n+\delta)k_t\right]\mathrm{d}t. \tag{31}$$

Note that capital accumulation remains driven by a deterministic differential equation conditional on (A_t, k_t, c_t): stochasticity enters only through the productivity term A_t.

Remark 4.1. *A natural extension would model capital depreciation as itself stochastic, replacing $(n+\delta)k_t\,\mathrm{d}t$ with $(n+\delta)k_t\,\mathrm{d}t + \sigma_k k_t\,\mathrm{d}\tilde{W}_t$ for an independent Brownian motion $\tilde{W}_t$. We leave this for future work to maintain tractability.*

5 Stochastic Dynamic Programming

5.1 The Value Function and HJB Equation

Let the value function $V : \mathbb{R}_{++} \times \mathbb{R}_{++} \to \mathbb{R}$ be defined by:

$$V(k, A) = \sup_{\{c_t, \ell_t\}} \mathbb{E}\left[\int_0^\infty e^{-\rho t}\big(\log c_t + \theta \log \ell_t\big)\, \mathrm{d}t \;\middle|\; k_0 = k,\ A_0 = A\right]. \tag{32}$$

By the principle of dynamic programming and Itô's lemma applied to $e^{-\rho t}V(k_t, A_t)$, the value function satisfies the Hamilton–Jacobi–Bellman (HJB) equation:

$$\boxed{\rho V = \max_{c,\ell} \left\{\log c + \theta \log \ell + V_k\big[Ak^\alpha(1-\ell)^{1-\alpha} - c - (n+\delta)k\big] + \mu A V_A + \tfrac{1}{2}\sigma^2 A^2 V_{AA}\right\},} \tag{33}$$

where $V_k = \partial V/\partial k$, $V_A = \partial V/\partial A$, $V_{AA} = \partial^2 V/\partial A^2$.

5.2 Optimality Conditions and Policy Functions

The maximization in (33) over (c, ℓ) yields:

$$c^*(k, A) = \frac{1}{V_k(k, A)}, \tag{34}$$

$$\frac{\theta}{\ell^*(k, A)} = V_k(k, A) \cdot (1-\alpha)Ak^\alpha(1-\ell^*)^{-\alpha}. \tag{35}$$

Substituting (34)–(35) into (33) and using $\log c^* = -\log V_k$ and $V_k(-c^*) = -1$, the *reduced* HJB equation becomes:

$$\rho V = -\log V_k - 1 + \theta \log \ell^* + V_k\big[Ak^\alpha(1-\ell^*)^{1-\alpha} - (n+\delta)k\big] + \mu A V_A + \tfrac{1}{2}\sigma^2 A^2 V_{AA}, \tag{36}$$

where ℓ^* is now understood as the policy function $\ell^*(k, A)$ satisfying (35).

Definition 5.1 (Viscosity Solution)**.** *A continuous function $V : \mathbb{R}^2_{++} \to \mathbb{R}$ is a* viscosity solution *of* (33) *if it is simultaneously a viscosity subsolution and supersolution in the sense of Crandall–Lions.*

Theorem 5.1 (Existence and Uniqueness)**.** *Under Assumption 3.1 and the GBM specification* (28)*, the HJB equation* (33) *admits a unique viscosity solution $V \in C(\mathbb{R}^2_{++})$, which equals the value function defined in* (32)*.*

Proof sketch. Standard results in viscosity solution theory (Crandall, Ishii, and Lions, 1992) apply once we verify: (i) the Hamiltonian is continuous and proper; (ii) the stochastic process (k_t, A_t) is non-degenerate on $\mathbb{R}^2_{++}$ (ensured by the GBM for A_t and strict positivity of capital under the policy constraint $c < y$); (iii) a comparison principle holds, which follows from the monotonicity of V_k and the strict concavity of the flow utility. □

Economic Intuition. The stochastic HJB equation generalizes the deterministic Euler equation: rather than a simple consumption-growth condition, the planner must now also account for the "option value" of waiting when productivity is uncertain. The term $\frac{1}{2}\sigma^2 A^2 V_{AA}$ is a Jensen's inequality correction: because the value function is concave in A (risk aversion), higher volatility reduces the certainty-equivalent productivity level.

6 Spectral Analysis of the Equilibrium Operator

6.1 The Bellman Operator

Define the Bellman operator $\mathcal{T} : C_b(\mathbb{R}^2_{++}) \to C_b(\mathbb{R}^2_{++})$ by:

$$(\mathcal{T}V)(k, A) = \max_{c,\ell} \left\{F(c, \ell) + V_k\big[Ak^\alpha(1-\ell)^{1-\alpha} - c - (n+\delta)k\big] + \mu A V_A + \tfrac{1}{2}\sigma^2 A^2 V_{AA}\right\}, \tag{37}$$

where $F(c, \ell) = \log c + \theta \log \ell$. The value function V^* is the fixed point: $\mathcal{T}V^* = \rho V^*$.

6.2 Fréchet Derivative of the Bellman Operator

Let V^* denote the equilibrium value function and consider a perturbation $V = V^* + \varepsilon u$ for $u \in C^2(\mathbb{R}^2_{++})$ and $\varepsilon \to 0$. The Fréchet derivative of $\mathcal{T}$ at V^* in direction u is:

$$D\mathcal{T}(V^*)u = \lim_{\varepsilon \to 0} \frac{\mathcal{T}(V^* + \varepsilon u) - \mathcal{T}(V^*)}{\varepsilon}. \tag{38}$$

Proposition 6.1. *Let (c^*, ℓ^*) be the optimal controls at V^*. Then:*

$$D\mathcal{T}(V^*)u = u_k \cdot g(k, A) + \mu A u_A + \tfrac{1}{2}\sigma^2 A^2 u_{AA}, \tag{39}$$

where $g(k, A) = Ak^\alpha(1 - \ell^*)^{1-\alpha} - c^* - (n + \delta)k$ is the net capital accumulation rate at the optimum.

Proof. Expanding $\mathcal{T}(V^* + \varepsilon u)$ and using the envelope theorem (since (c^*, ℓ^*) already maximize at V^*, first-order effects of perturbing the control vanish), the leading order perturbation comes entirely from the direct dependence of $\mathcal{T}$ on V. Differentiating under the operator yields (39). □

Proposition 6.2 (Linearized Dynamics)**.** *The linearized equilibrium dynamics around V^* are governed by the differential operator:*

$$\mathcal{L}u = g(k, A)\, u_k + \mu A\, u_A + \tfrac{1}{2}\sigma^2 A^2 u_{AA}. \tag{40}$$

Theorem 6.3 (Local Stability)**.** *If the spectral radius of $D\mathcal{T}(V^*)$ (as an operator on $L^2(\Omega, \mu)$ for the stationary measure μ) satisfies $\varrho(D\mathcal{T}(V^*)) < 1$, then the equilibrium V^* is locally asymptotically stable under the iteration $V_{n+1} = \mathcal{T}V_n$.*

7 Resolvent and Spectral Gap Analysis

7.1 Generator and Resolvent

The operator $\mathcal{L}$ defined in (40) is the *infinitesimal generator* of the diffusion process (k_t, A_t) under the optimal policy. Define the resolvent:

$$\mathcal{R}(\lambda) = (\lambda\mathcal{I} - \mathcal{L})^{-1}, \qquad \mathrm{Re}(\lambda) > 0. \tag{41}$$

Theorem 7.1 (Exponential Convergence)**.** *Suppose $\mathcal{L}$ generates a strongly continuous semigroup $\{P_t\}_{t\geq 0}$ on $L^2(\mathbb{R}^2_{++}, \mu)$ where μ is the stationary distribution. If the leading non-zero eigenvalue λ_1 of $\mathcal{L}$ satisfies $\mathrm{Re}(\lambda_1) < 0$, then for any $u_0 \in L^2(\mathbb{R}^2_{++}, \mu)$:*

$$\|P_t u_0 - \Pi u_0\|_{L^2(\mu)} \leq C e^{\mathrm{Re}(\lambda_1)t} \|u_0\|_{L^2(\mu)}, \tag{42}$$

where $\Pi u_0 = \int u_0 \,\mathrm{d}\mu$ is the projection onto the stationary state.

Proof. By spectral decomposition, $P_t u_0 = \sum_i e^{\lambda_i t}\langle u_0, \phi_i\rangle \phi_i$, where $\{\phi_i\}$ are the eigenfunctions of $\mathcal{L}$. Since $\lambda_0 = 0$ corresponds to the stationary distribution and $\mathrm{Re}(\lambda_i) \leq \mathrm{Re}(\lambda_1) < 0$ for all $i \geq 1$, the non-stationary components decay at rate $e^{\mathrm{Re}(\lambda_1)t}$, giving the stated bound. □

Economic Intuition. The spectral gap $\gamma = |\mathrm{Re}(\lambda_1)|$ is the key macroeconomic persistence parameter: it measures how quickly the economy "forgets" initial conditions after a shock. A small γ (eigenvalue near zero) implies slow mean reversion—shocks are long-lived, matching the empirical persistence of business cycles. A large γ implies rapid convergence, consistent with economies near a stable steady state.

8 Diffusion Geometry and the Witten Laplacian

8.1 Potential Landscape and Witten Transform

Define the macro potential function:

$$U(k, A) = -\log V(k, A). \tag{43}$$

Since $V > 0$ (the value of living is positive under our assumptions), U is well-defined and finite on $\mathbb{R}^2_{++}$.

The generator $\mathcal{L}$ can be written in divergence form. Define the Witten-transformed operator:

$$\widetilde{\mathcal{L}} = e^{-U/2}\mathcal{L}e^{U/2}. \tag{44}$$

Proposition 8.1 (Witten Laplacian Structure)**.** *The transformed operator $\widetilde{\mathcal{L}}$ is self-adjoint on $L^2(\mathbb{R}^2_{++})$ and takes the form:*

$$\widetilde{\mathcal{L}}f = \mathcal{L}f + \nabla U \cdot \nabla f + (\tfrac{1}{2}\mathcal{L}U + \tfrac{1}{4}|\nabla U|^2)f, \tag{45}$$

acting as a Schrödinger-type operator with potential $\Psi = -\frac{1}{2}\mathcal{L}U - \frac{1}{4}|\nabla U|^2$.

Proposition 8.2 (Equilibrium Fluctuations as Diffusion)**.** *Macroeconomic equilibrium fluctuations correspond to diffusion on the potential landscape $U(k, A) = -\log V(k, A)$. Potential wells of U (i.e., local maxima of V) attract the diffusion, corresponding to stable macroeconomic configurations. Saddle points of V correspond to transition paths between regimes.*

9 Morse Landscape of Macroeconomic Equilibria

9.1 Critical Point Structure

Define the macro potential:

$$\Phi(k, A) = -V(k, A). \tag{46}$$

Critical points of Φ (equivalently, of V) satisfy:

$$\nabla V(k, A) = \mathbf{0}. \tag{47}$$

Definition 9.1 (Morse Value Function)**.** *The value function V is called* Morse *if at every critical point $p = (k^*, A^*)$, the Hessian $\mathrm{Hess}(V)\big|_p$ is non-singular (all eigenvalues are non-zero).*

Theorem 9.1 (Topological Characterization of Equilibria)**.** *If V is a Morse function on $\mathbb{R}^2_{++}$, then:*

(i) The number of macroeconomic equilibria (critical points) is finite.

(ii) Each equilibrium has a well-defined Morse index $\mu_p \in \{0, 1, 2\}$: index-0 points are local maxima of V (stable equilibria), index-1 points are saddles (transition states), and index-2 points are local minima (unstable equilibria).

(iii) The Euler characteristic of the state space satisfies $\chi = \sum_p (-1)^{\mu_p}$.

Proof. Statement (i) follows from compactness of level sets under the growth conditions on V. Statements (ii) and (iii) are the content of the Morse inequalities applied to $-V$. □

Economic Intuition. The Morse index of an equilibrium tells us its stability type. A Morse index of 0 (maximum of V) corresponds to a locally stable growth regime—the economy converges to it from nearby states. A Morse index of 1 (saddle of V) is a knife-edge: the economy converges from one direction but diverges in another, analogous to a development trap in the growth literature. This provides a rigorous topological foundation for multi-stability in macroeconomic models.

10 Spectral Asymptotics and Metastable Regimes

10.1 The Setting

The diffusion generator $\mathcal{L}$ (equation (40)) induces a stochastic semigroup $P_t = e^{t\mathcal{L}}$. Long-run dynamics are governed by the low-lying eigenvalues of $\mathcal{L}$ (equivalently, of $\widetilde{\mathcal{L}}$). In the small-noise limit $\sigma \to 0$, these eigenvalues become exponentially small, associated with rare transitions between macroeconomic regimes.

10.2 Witten-Type Spectral Asymptotics

Theorem 10.1 (Macro Witten Spectral Asymptotics)**.** *Let $U = -\log V$ have finitely many non-degenerate local minima $\{p_i\}_{i=1}^m$ and suppose the potential barriers between them are:*

$$\Delta U_i = \min_{path\ \gamma_i} \max_s U(\gamma_i(s)) - U(p_i), \qquad i = 1, \ldots, m-1. \tag{48}$$

Then as $\sigma \to 0$, the i-th smallest positive eigenvalue of $\widetilde{\mathcal{L}}$ satisfies:

$$\lambda_i \sim C_i \exp\left(-\frac{2\Delta U_i}{\sigma^2}\right), \tag{49}$$

where $C_i > 0$ depends on local geometry at the saddle and minima (specifically, ratios of Hessian determinants at these points).

Proof sketch. This follows from Helffer–Sjöstrand theory for Witten Laplacians in the semiclassical limit. The key step is showing that the low-lying eigenvalues of $\widetilde{\mathcal{L}}$ correspond to quasi-modes concentrated near the potential minima $\{p_i\}$, and that tunneling between wells contributes exponentially small eigenvalues via WKB analysis along the instanton paths $\{\gamma_i\}$. □

10.3 Economic Interpretation

10.3.1 Regime-Switching Timescales

The spectral gap $\gamma = |\lambda_1|$ directly governs the timescale of macroeconomic regime transitions. By the Kramers formula, the expected transition time from regime i to regime j satisfies:

$$\mathbb{E}[\tau_{i\to j}] \approx C_{ij}^{-1} \exp\left(\frac{2\Delta U_{ij}}{\sigma^2}\right). \tag{50}$$

10.3.2 Interpretation of Regimes

The potential wells of U correspond to persistent macroeconomic configurations:

- **High-capital growth regimes:** deep wells with large ΔU—the economy persists in this state for long periods.
- **Stagnation traps:** local minima of V with intermediate ΔU—escape requires a sufficiently large positive shock.
- **Transition states (saddles):** unstable configurations between regimes; the economy passes through these only briefly.

10.3.3 Return-Time Distributions

Let τ denote the first return time to a neighborhood of the equilibrium. Spectral theory implies:

$$\mathbb{P}(\tau > t) \approx e^{-\gamma t}, \qquad t \gg \gamma^{-1}. \tag{51}$$

This exponential tail arises from the spectral gap and implies that macroeconomic recessions and recoveries should exhibit exponential duration distributions—a testable prediction consistent with empirical evidence on business cycle durations.

Algorithm 1 Spectral Macro Regime Detection

Require: Stochastic macro paths $\{(k_t, A_t)\}_{t=0}^T$, basis functions $\{\phi_j\}_{j=1}^J$

1: Estimate the generator matrix $\hat{L}_{ij} = \langle \phi_i, \mathcal{L}\phi_j \rangle$ using time-averages.
2: Compute the leading m eigenvalues $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_m$ of $\hat{L}$.
3: Identify spectral gaps: $\hat{\gamma}_i = |\hat{\lambda}_i - \hat{\lambda}_{i+1}|$.
4: Assign regime memberships via dominant eigenvectors (fuzzy k-means on eigenvectors).
5: Estimate transition rates as $\hat{\lambda}_i$ and characteristic timescales as $|\hat{\lambda}_i|^{-1}$.

Ensure: Regime labels $\{R_t\}_{t=0}^T$, estimated spectral gaps $\{\hat{\gamma}_i\}$.

11 Neural PDE Solver for the HJB Equation

11.1 Motivation and Architecture

The HJB equation (33) is a nonlinear second-order PDE on the two-dimensional state space $(k, A) \in \mathbb{R}^2_{++}$. Classical numerical methods (finite differences, finite elements) suffer from the curse of dimensionality when extended to higher-dimensional models. We propose a neural network approximation:

$$V_\phi(k, A) \approx V(k, A), \qquad \phi \in \mathbb{R}^P, \tag{52}$$

where V_ϕ is a deep neural network with parameters ϕ, representing the HJB value function.

We employ a *physics-informed* architecture: the network takes (k, A) as input and is trained to satisfy the HJB equation residual. Specifically, we parametrize:

$$V_\phi(k, A) = V_{\text{base}}(k, A) + \text{NN}_\phi(k, A), \tag{53}$$

where V_{base} is an analytically derived baseline (e.g., the deterministic Ramsey value function) and NN_ϕ is a correction network. This accelerates convergence by initializing near the true solution.

11.2 Loss Function and Training

Define the HJB residual at (k, A) under policy (c^*_ϕ, ℓ^*_ϕ) derived from V_ϕ:

$$\mathcal{R}_\phi(k, A) = \rho V_\phi - \log c^*_\phi - \theta \log \ell^*_\phi \\ - \partial_k V_\phi \big[Ak^\alpha (1 - \ell^*_\phi)^{1-\alpha} - c^*_\phi - (n + \delta)k\big] - \mu A \partial_A V_\phi - \tfrac{1}{2}\sigma^2 A^2 \partial_{AA} V_\phi. \quad (54)$$

The training loss combines the HJB residual with boundary and regularity penalties:

$$\mathcal{L}(\phi) = \underbrace{\mathbb{E}_{(k,A)\sim\nu}\big[\mathcal{R}_\phi(k, A)^2\big]}_{\text{HJB residual}} + \beta_1 \underbrace{\mathcal{L}_{\text{BC}}(\phi)}_{\text{boundary}} + \beta_2 \underbrace{\mathcal{L}_{\text{TV}}(\phi)}_{\text{regularity}}, \quad (55)$$

where ν is a sampling distribution over the state space (e.g., a truncated log-normal fitted to observed (k, A) pairs), $\beta_1, \beta_2 > 0$ are penalty weights, $\mathcal{L}_{\text{BC}}$ enforces transversality and Inada-condition-consistent boundary behavior, and $\mathcal{L}_{\text{TV}}$ is a total-variation regularizer promoting smooth solutions.

Theorem 11.1 (Approximation Error Bound)**.** *Let V^* be the true viscosity solution to (33) and V_{ϕ^*} be the minimizer of (55). Assume $V^* \in H^2(\mathcal{D})$ for a compact domain $\mathcal{D} \subset \mathbb{R}^2_{++}$. Then for a ReLU network with L layers and width W:*

$$\|V_{\phi^*} - V^*\|_{L^2(\mathcal{D})} \le C_{approx} W^{-2/d} L^{-2/d} + C_{opt}\sqrt{\mathcal{L}(\phi^*)}, \quad (56)$$

where $d = 2$ is the state-space dimension and C_{approx}, C_{opt} are constants depending on V^* and the loss structure.

Proof sketch. The approximation term $C_{\text{approx}} W^{-2/d}$ follows from universal approximation theory for Sobolev functions (DeVore et al., 1989). The optimization term $C_{\text{opt}}\sqrt{\mathcal{L}(\phi^*)}$ follows from the Gårding inequality applied to the second-order elliptic operator defined by $\mathcal{L}$, which gives coercivity of the HJB bilinear form and thus an H^1 control on the approximation error in terms of the residual. □

Algorithm 2 Neural PDE Training for the HJB Equation

Require: Network architecture NN_ϕ, baseline V_{base}, hyperparameters β_1, β_2, η
Initialize: $\phi_0 \leftarrow$ Xavier initialization; $V_\phi = V_{\text{base}} + \text{NN}_\phi$
for epoch $= 1, \ldots, N_{\text{epoch}}$ **do**
  Sample mini-batch $\{(k_i, A_i)\}_{i=1}^B \sim \nu$
  Compute optimal controls: $c^*_\phi = 1/\partial_k V_\phi$, ℓ^*_ϕ from (35)
  Evaluate HJB residuals $\{\mathcal{R}_\phi(k_i, A_i)\}$ using automatic differentiation
  Compute loss $\hat{\mathcal{L}}(\phi) = B^{-1}\sum_i \mathcal{R}_\phi(k_i, A_i)^2 + \beta_1\mathcal{L}_{\text{BC}} + \beta_2\mathcal{L}_{\text{TV}}$
  Update: $\phi \leftarrow \phi - \eta\nabla_\phi\hat{\mathcal{L}}(\phi)$ ▷ Adam optimizer
  if $\hat{\mathcal{L}}(\phi) < \varepsilon_{\text{tol}}$ **then break**
  end if
end for
Ensure: Approximate value function V_ϕ, policy functions c^*_ϕ, ℓ^*_ϕ

12 Econometric Estimation

12.1 Identification Strategy

The structural parameters $(\alpha, \theta, \sigma, \mu)$ can be identified from macroeconomic panel data. Taking logs of the per-capita production function (4):

$$\log y_t = \log A_t + \alpha \log k_t + (1 - \alpha)\log(1 - \ell_t). \quad (57)$$

Since $\log A_t$ follows an arithmetic Brownian motion (29), we can decompose:

$$\log A_t = \log A_0 + \big(\mu - \tfrac{1}{2}\sigma^2\big)\, t + \sigma W_t, \quad (58)$$

so $\log A_t - \log A_{t-1} = (\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma\Delta W_t$, which is i.i.d. $\mathcal{N}((\mu - \frac{1}{2}\sigma^2)\Delta t,\ \sigma^2\Delta t)$. This Gaussianity enables maximum likelihood estimation.

Parameter	Description	Identification	Data Source	Method
α	Capital income share	Factor shares	National accounts	OLS regression
A_t	TFP level	Solow residual	Penn World Tables	HP/BK filtering
θ	Leisure preference	Labor supply elasticity	Time-use surveys	GMM calibration
μ	TFP drift	Mean TFP growth	Long-run averages	MLE
σ	TFP volatility	Variance of $\Delta \log A_t$	GDP deflated series	MLE / GARCH
ρ	Discount rate	Euler equation moment	Consumption data	SMM

Table 1: Structural parameter identification strategy.

12.2 Moment Conditions

The model implies the following testable moment conditions, which can be used for GMM estimation:

$$\mathbb{E}\left[\frac{\dot{c}_t}{c_t} - (r_t - \rho - \delta)\right] = 0, \quad \text{(Euler equation)} \tag{59}$$

$$\mathbb{E}\left[\ell_t - \frac{\theta}{\theta + 1 - \alpha}\right] = 0, \quad \text{(leisure optimality)} \tag{60}$$

$$\mathbb{E}\left[\log(A_t/A_{t-1}) - \left(\mu - \tfrac{1}{2}\sigma^2\right)\Delta t\right] = 0. \quad \text{(TFP dynamics)} \tag{61}$$

13 Machine Learning Surrogate Model

13.1 Motivation

Solving the HJB equation for each parameter vector $(\alpha, \theta, \sigma, \mu, \rho)$ is computationally expensive. For policy analysis, counterfactual exercises, and Bayesian estimation, one needs thousands of model solutions. We train a *surrogate model*—a neural network that directly maps parameters and state variables to equilibrium outcomes.

13.2 Surrogate Architecture and Training

Define the surrogate mapping:

$$f_\phi : \underbrace{(A, \alpha, \theta, k)}_{\text{inputs}} \mapsto \underbrace{(y, w, r, \ell^*, V)}_{\text{outputs}}, \tag{62}$$

where f_ϕ is a neural network with parameters ϕ. The training dataset $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$ is generated by solving the model numerically for a Latin hypercube sample of parameter vectors.

The surrogate loss combines prediction error with *equilibrium consistency* constraints:

$$\mathcal{L}_{\text{surr}}(\phi) = \sum_{i=1}^N \|f_\phi(\mathbf{x}_i) - \mathbf{y}_i\|^2 + \gamma \sum_{i=1}^N \mathcal{C}(f_\phi(\mathbf{x}_i), \mathbf{x}_i), \tag{63}$$

where $\mathcal{C}$ penalizes violations of the equilibrium conditions (firm FOCs, household optimality, goods market clearing), ensuring that the surrogate's outputs are internally consistent even at interpolated parameter values.

Theorem 13.1 (Surrogate Generalization Bound)**.** *Let f_{ϕ^*} minimize $\mathcal{L}_{surr}$ over a hypothesis class $\mathcal{F}$ of Rademacher complexity $\mathcal{R}_N(\mathcal{F})$. With probability $1 - \delta$:*

$$\mathbb{E}\left[\|f_{\phi^*}(\mathbf{x}) - \mathbf{y}\|^2\right] \leq \hat{\mathcal{L}}_{surr}(\phi^*) + 2\mathcal{R}_N(\mathcal{F}) + 3\sqrt{\frac{\log(2/\delta)}{2N}}. \tag{64}$$

Proof. Standard application of Rademacher complexity bounds (Bartlett and Mendelson, 2002) to the squared loss, combined with uniform convergence over $\mathcal{F}$. □

Algorithm 3 Equilibrium-Constrained Surrogate Training

Require: Parameter ranges Θ, number of training samples N, penalty weight γ
Data generation: Sample $\{(\alpha^{(i)}, \theta^{(i)}, \sigma^{(i)}, \mu^{(i)}, k^{(i)}, A^{(i)})\}_{i=1}^{N}$ via Latin hypercube.
for $i = 1, \ldots, N$ **do**
    Solve HJB numerically (via Algorithm 2) to obtain $(y^{(i)}, w^{(i)}, r^{(i)}, \ell^{*(i)}, V^{(i)})$.
    Store $(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})$ in dataset $\mathcal{D}$.
end for
Split $\mathcal{D}$ into train (80%), validation (10%), test (10%) sets.
Initialize network f_ϕ with L layers, width W, residual connections.
for epoch $= 1, \ldots, N_{\text{epoch}}$ **do**
    Sample mini-batch; compute prediction loss and equilibrium penalty.
    Update ϕ via Adam with cosine learning rate schedule.
    Monitor validation loss; apply early stopping.
end for
Validate on test set: report R^2, RMSE, and maximum equilibrium-condition violation.
Ensure: Trained surrogate f_{ϕ^*} with documented error bounds.

Remark 13.1 (Uncertainty Quantification)**.** *Deploying the surrogate for policy analysis requires quantifying its uncertainty. We recommend training an ensemble of M surrogates $\{f_{\phi_m}\}_{m=1}^{M}$ with different random seeds and using the ensemble variance $\mathrm{Var}_m[f_{\phi_m}(\mathbf{x})]$ as a measure of epistemic uncertainty. Parameter regions with high uncertainty can then be targeted for additional HJB solves to augment the training set.*

14 Multi-Country Network Equilibrium

14.1 Model with International Capital Flows

Consider M economies indexed by $i = 1, \ldots, M$. Each economy has its own TFP A_i, capital k_i, population growth n_i, and preference parameters, but shares the same production technology. Economies are linked by bilateral capital flows described by the matrix $W \in \mathbb{R}^{M \times M}$, where $W_{ij} \geq 0$ is the rate of capital inflow from country j to country i (with $W_{ii} = 0$).

Capital dynamics for country i:

$$\dot{k}_i = A_i k_i^{\alpha}(1 - \ell_i^*)^{1-\alpha} - c_i - (n_i + \delta)k_i + \sum_{j \neq i} W_{ij} k_j - \sum_{j \neq i} W_{ji} k_i. \tag{65}$$

In vector form, with $\mathbf{k} = (k_1, \ldots, k_M)^\top$ and $F_i(\mathbf{k}) = A_i k_i^{\alpha}(1 - \ell_i^*)^{1-\alpha} - c_i - (n_i + \delta)k_i$:

$$\dot{\mathbf{k}} = F(\mathbf{k}) + \tilde{W}\mathbf{k}, \tag{66}$$

where $\tilde{W}_{ii} = -\sum_{j \neq i} W_{ji}$ (net outflow) and $\tilde{W}_{ij} = W_{ij}$ for $j \neq i$.

Proposition 14.1 (Global Equilibrium)**.** *The global equilibrium $\mathbf{k}^*$ satisfies $F(\mathbf{k}^*) + \tilde{W}\mathbf{k}^* = \mathbf{0}$.*

Theorem 14.2 (Multi-Country Stability)**.** *Let $J = DF(\mathbf{k}^*) + \tilde{W}$ be the Jacobian of the right-hand side of (66) at $\mathbf{k}^*$. The global equilibrium is locally asymptotically stable if and only if all eigenvalues of J have strictly negative real parts.*

Proof. Local stability follows directly from the Hartman–Grobman theorem: the nonlinear system (66) is topologically equivalent to its linearization near $\mathbf{k}^*$, and the linearized system is asymptotically stable iff all eigenvalues of J satisfy $\mathrm{Re}(\lambda_i) < 0$. □

Economic Intuition. The matrix $\tilde{W}$ introduces off-diagonal terms into the Jacobian, reflecting the transmission of shocks across borders. Even if each country individually has a stable equilibrium ($J^{(i)} < 0$), cross-country capital flows can generate instability if $\tilde{W}$ has a sufficiently large positive eigenvalue. This formalizes the notion of *contagion*: a financial crisis in one country can destabilize a globally stable system.

15 Conclusion

This paper has developed a comprehensive macroeconomic framework rooted in the simple but powerful observation that population scaling of capital and leisure yields a tractable per-capita equilibrium. Starting

from a static closed-form solution (13) that cleanly separates preference, technology, and policy effects, we have extended the model along four complementary dimensions.

Dynamic extension. The Ramsey model yields the Euler equation (23), $\dot{c}/c = r - (\rho + n + \delta)$, and establishes saddle-path stability via a negative Jacobian determinant. The steady-state capital stock satisfies the modified golden rule $r^* = \rho + n + \delta$, equating the marginal product of capital with the full cost of capital accumulation.

Stochastic extension. Geometric Brownian motion productivity shocks lead to the HJB equation (33), whose viscosity solution characterizes optimal behavior under uncertainty. The key new insight is the Jensen's inequality correction $\frac{1}{2}\sigma^2 A^2 V_{AA}$, which compresses effective productivity for risk-averse planners.

Spectral-geometric extension. The generator $\mathcal{L}$ of the equilibrium diffusion has a rich spectral structure. The spectral gap γ governs the speed of equilibration, and Witten-type asymptotics (49) reveal that macroeconomic regime-switching timescales are exponentially sensitive to productivity volatility σ. Morse theory provides a complete topological classification of multi-stability.

Computational extension. Neural PDE solvers and ML surrogates provide scalable tools for computing and approximating equilibria, with formal approximation error bounds (Theorems 11.1 and 13.1). The equilibrium-consistency penalty in the surrogate loss (63) is a key innovation ensuring economically coherent predictions at interpolated parameters.

Multi-country extension. The network equilibrium (Section 14) shows that global stability depends on the full Jacobian $J = DF + \tilde{W}$, incorporating both domestic dynamics and international capital flow transmission.

15.1 Future Directions

Several extensions present themselves naturally. First, incorporating heterogeneous agents would break the representative-agent assumption and introduce distributional dynamics alongside the aggregate ones. Second, endogenous labor supply in the network model raises questions about wage equalization and optimal capital controls. Third, the surrogate model could be deployed within a Bayesian sequential Monte Carlo framework for real-time parameter estimation from streaming macroeconomic data. Finally, an important open question is whether the Morse and Witten machinery extends to infinite-dimensional state spaces, as would arise in models with capital vintage structure or demographic heterogeneity.

A Proof of the Intratemporal Optimality Condition

We provide a self-contained derivation of (13). The household's problem in the static model is:

$$\max_{\ell \in (0,1)} \log\big[Ak^\alpha (1-\ell)^{1-\alpha}\big] + \theta \log \ell,$$

using goods market clearing $c = y$. Differentiating with respect to ℓ:

$$\frac{-(1-\alpha)}{1-\ell} + \frac{\theta}{\ell} = 0,$$

which gives $\theta(1-\ell) = \ell(1-\alpha)$ and thus $\ell^* = \theta/(\theta + 1 - \alpha)$. The second-order condition is:

$$-\frac{(1-\alpha)}{(1-\ell)^2} - \frac{\theta}{\ell^2} < 0,$$

which is strictly negative for all $\ell \in (0,1)$, confirming a global maximum.

B References

[1] Acemoglu, D. (2008). *Introduction to Modern Economic Growth*. Princeton University Press.

[2] Arrow, K.J. and Debreu, G. (1954). Existence of an equilibrium for a competitive economy. *Econometrica*, 22(3), 265–290.

[3] Bartlett, P.L. and Mendelson, S. (2002). Rademacher and Gaussian complexities: Risk bounds and structural results. *Journal of Machine Learning Research*, 3, 463–482.

[4] Crandall, M.G., Ishii, H., and Lions, P.-L. (1992). User's guide to viscosity solutions of second order partial differential equations. *Bulletin of the AMS*, 27(1), 1–67.

[5] Duffie, D. and Epstein, L. (1992). Stochastic differential utility. *Econometrica*, 60(2), 353–394.

[6] Helffer, B. and Sjöstrand, J. (1984). Multiple wells in the semi-classical limit I. *Communications in Partial Differential Equations*, 9(4), 337–408.

[7] Lucas, R.E. (1988). On the mechanics of economic development. *Journal of Monetary Economics*, 22(1), 3–42.

[8] Ramsey, F.P. (1928). A mathematical theory of saving. *Economic Journal*, 38(152), 543–559.

[9] Romer, P.M. (1986). Increasing returns and long-run growth. *Journal of Political Economy*, 94(5), 1002–1037.

[10] Stokey, N.L. and Lucas, R.E. (1989). *Recursive Methods in Economic Dynamics*. Harvard University Press.

[11] Witten, E. (1982). Supersymmetry and Morse theory. *Journal of Differential Geometry*, 17(4), 661–692.

C Glossary of Key Notation

Symbol	**Meaning**
N	Population size
k, K	Per-capita capital; aggregate capital $K = kN$
ℓ, L	Per-capita leisure; aggregate leisure $L = \ell N$
c	Per-capita consumption
y	Per-capita output
α	Capital income share
θ	Leisure preference parameter
ρ	Household discount rate
n	Population growth rate
δ	Capital depreciation rate
A	Total factor productivity
μ, σ	TFP drift and volatility (GBM)
$V(k, A)$	Stochastic value function
$\mathcal{L}$	Infinitesimal generator of equilibrium diffusion
λ_i	i-th eigenvalue of $\mathcal{L}$
$\gamma = \lvert\lambda_1\rvert$	Spectral gap (regime-switching rate)
$U = -\log V$	Macro potential function
ΔU_i	Potential barrier between macroeconomic regimes
$W, \tilde{W}$	International capital flow matrix
$J = DF + \tilde{W}$	Network equilibrium Jacobian

The End